Adarsh Dwivedi

AI Product Engineer

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Final-year CS engineer who builds AI products end to end, from the evaluation harness to the surface a user touches. Shipped public products, 38 repositories in all, across 10+ national hackathons and datathons, each solving a stated real-world problem for police analysts, farmers and regulators. At Deloitte, migrated a production LangGraph agent onto a self-hosted Aegra server and proved it on a live SAP tenant, 8 of 8 tasks with human approval intact. Treats latency, cost and failure behaviour as product decisions, not implementation details.

EXPERIENCE

Deloitte India, Digital Excellence Centre **Jun 2026 - Aug 2026**

AI Engineer (Agentic AI) Intern Gurugram, India

Migrated SAP Agentic, Deloitte's system that automates SAP configuration, from LangGraph running inside a FastAPI request handler onto a self-hosted Aegra agent server, with threads, runs and checkpoints in PostgreSQL inside the firewall at no licence cost.

Proved the new path on the live SAP tenant: a Bank Accounting goal was planned into 8 configuration tasks, executed in a real browser through Playwright MCP, recovered from a duplicate-entry conflict and a mis-filled field, and finished 8 of 8 in about 15 minutes.

Replaced request-time graph assembly with a graph factory that registers one static graph and carries per-request variation in run state, verifying the migrated output identical to the original field by field behind an unchanged API.

PROJECTS

AGENTIQ, Agentic API Testing and Security Validation *(B.Tech final-year project)* | **Live** | **GitHub** Node.js, MCP, LLM Agents, Zod

Functional tests lived in Postman, security tests in ZAP, and nobody could account for what an LLM did against a live endpoint. Built one platform generating both from a URL and plain-English intent.

Made autonomy safe by construction, not by prompt: 21 Zod-validated tools across 6 risk classes, with per-host grants, a filesystem jail, a process sandbox and an SSRF guard on every agent action.

Probed 8 vulnerability families mapped to the OWASP API Top 10 against a benign baseline, so a finding needs a material deviation rather than a suspicious string. 719 tests gate every change in CI.

CacheLLM, Semantic Caching Proxy for LLM APIs *(Published on PyPI)* | **Live** | **GitHub** Python, FastAPI, Embeddings

Teams pay full price to answer the same question reworded. Shipped a drop-in proxy on PyPI keeping the OpenAI request and response shape including streaming, so adoption costs one base-URL change.

Matched a normalised hash first, then cosine similarity over unit-normalised embeddings, with the safe threshold measured per embedding model rather than assumed, and six providers behind one interface.

Vayu, National Air-Quality Intelligence and Enforcement Platform *(Deployed on Google Cloud Run)* | **Live** | **GitHub** LightGBM, CNN-LSTM, DuckDB

Enforcement targets the dirtiest sensor, not the site where action pays most. Forecast air quality with LightGBM on ground stations fused with a CNN-LSTM over a 15,360-cell satellite grid.

Ranked interventions by Gaussian-plume modelled return, then kept PyTorch out of the request path entirely, training offline and serving precomputed scores from one embedded DuckDB file on Cloud Run.

MedGuardX, Context-Aware PII and PHI Masking *(Published on PyPI)* | **Live** | **GitHub** Python, FastAPI, spaCy, JWT

Masking is not one operation, since a doctor, a billing clerk and a researcher need different views of one record. Built a role-by-purpose-by-consent matrix resolving to mask, anonymise, allow or deny.

Matched identifiers by format rather than by model, including Aadhaar with Verhoeff checksum validation, PAN, MRN, cards and IBAN, with overlaps resolved through a deterministic priority table.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript, TypeScript, SQL

AI and LLM Engineering: LLM Application Engineering, Agent Frameworks and MCP, Retrieval (RAG) and Vector Search, Embeddings, Evaluation Harnesses, Context Engineering, Guardrails and Sandboxing, Inference Cost and Latency Optimisation, Model Serving

Backend and APIs: FastAPI, Pydantic, Node.js, Express, REST APIs, PostgreSQL, MongoDB, DuckDB, Redis, Zod Schema Validation

Infrastructure and Tooling: Docker, AWS, Google Cloud Run, Vercel, Render, GitHub Actions, CI/CD, Linux, Git, Observability, Reproducible Builds

Product: Product Strategy, Product Discovery, Problem Framing, PRD and Technical Specification, Roadmapping, Prioritisation (RICE), User Research, Metrics and KPI Definition, Adoption and Engagement, A/B Testing, Stakeholder Management, Competitive Analysis, Agile and Scrum, Go-to-Market

ACHIEVEMENTS

Competed in 10+ national hackathons and datathons: KSP Datathon 2026 finalist, Level 2 with Kadi for the Karnataka State Police; AWS AI for Bharat finalist with VaidyaMitra; Hacksplosion 2026 with Deloitte India, clearing Levels 1, 2 and 3; and SWITCH Energy-X, 11th of 93.

HackerRank Orchestrate: ranked #12 with a silver medal, top 1% (Sep 2026), up from #290 with a bronze medal, top 15% (Aug 2026), in 24-hour agentic AI challenges.

micro1 Frontier Engineering Challenge 2026: built a research-artifact verifier measured across 742 repositories.

38 public repositories, most deployed and free to use, with three packages published to PyPI and every reported number reproducible from the repository.

EDUCATION

The LNM Institute of Information Technology, Jaipur **2023 - 2027**

B.Tech, Computer Science and Engineering CGPA: 7.11 / 10

LEADERSHIP

Coordinator, Debate Society, LNMIIT Jaipur (May 2025 - Apr 2026): led a 150+ member society across debates, discussions and inter-college competitions.

Secretariat and Organizer, LNMIIT Model United Nations (Aug 2025 - Jan 2026): organised a large-scale MUN conference, coordinating delegates, committees, schedules and logistics.